PHILIPPINE SOCIETY OF
ANIMAL NUTRITIONISTS

39th ANNUAL CONVENTION

Fueling Progress: Shaping Animal Nutrition for a Competitive and Sustainable Future

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Grand Ballroom,
Okada Manila









PHILIPPINE SOCIETY OF ANIMAL NUTRITIONISTS
39th Annual Convention | October 6, 2026

Ruminants' resilience in tropical production system

Integrated insights across heat stress and nutrition

Pietro Celi (DVM, PhD)
Murdoch University, Australia

Outline

Murdoch University

- Heat stress resilience under climate variability
- Heat stress and gut health
- Heat stress and glucose
- Feed efficiency and precision nutrition
- Conclusions

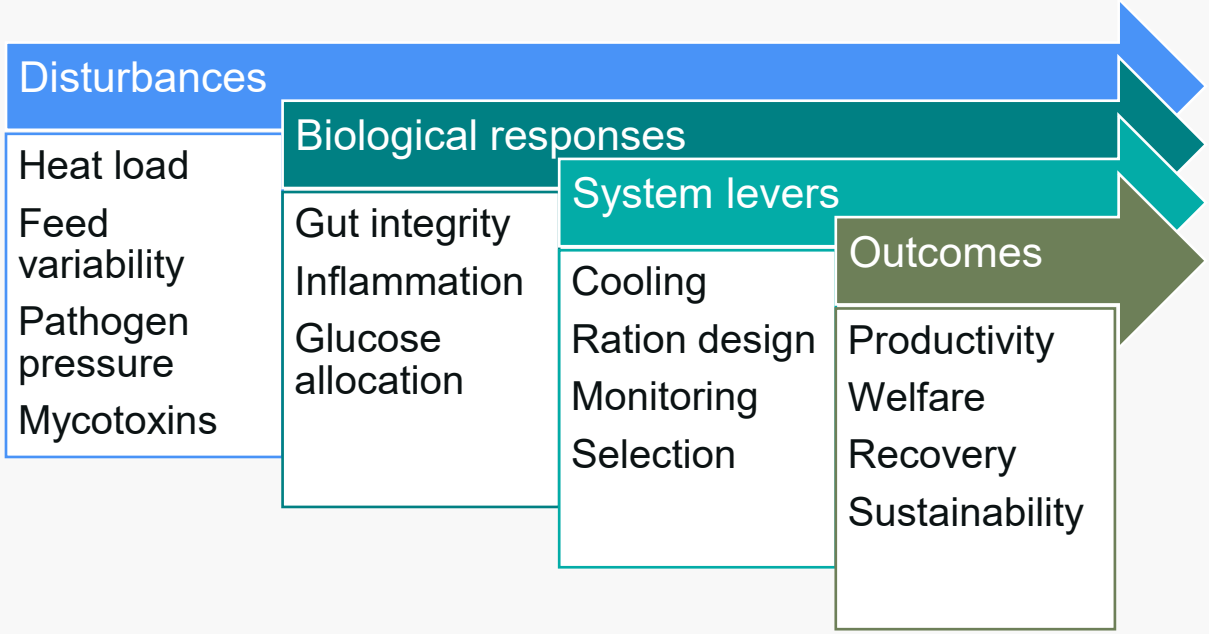


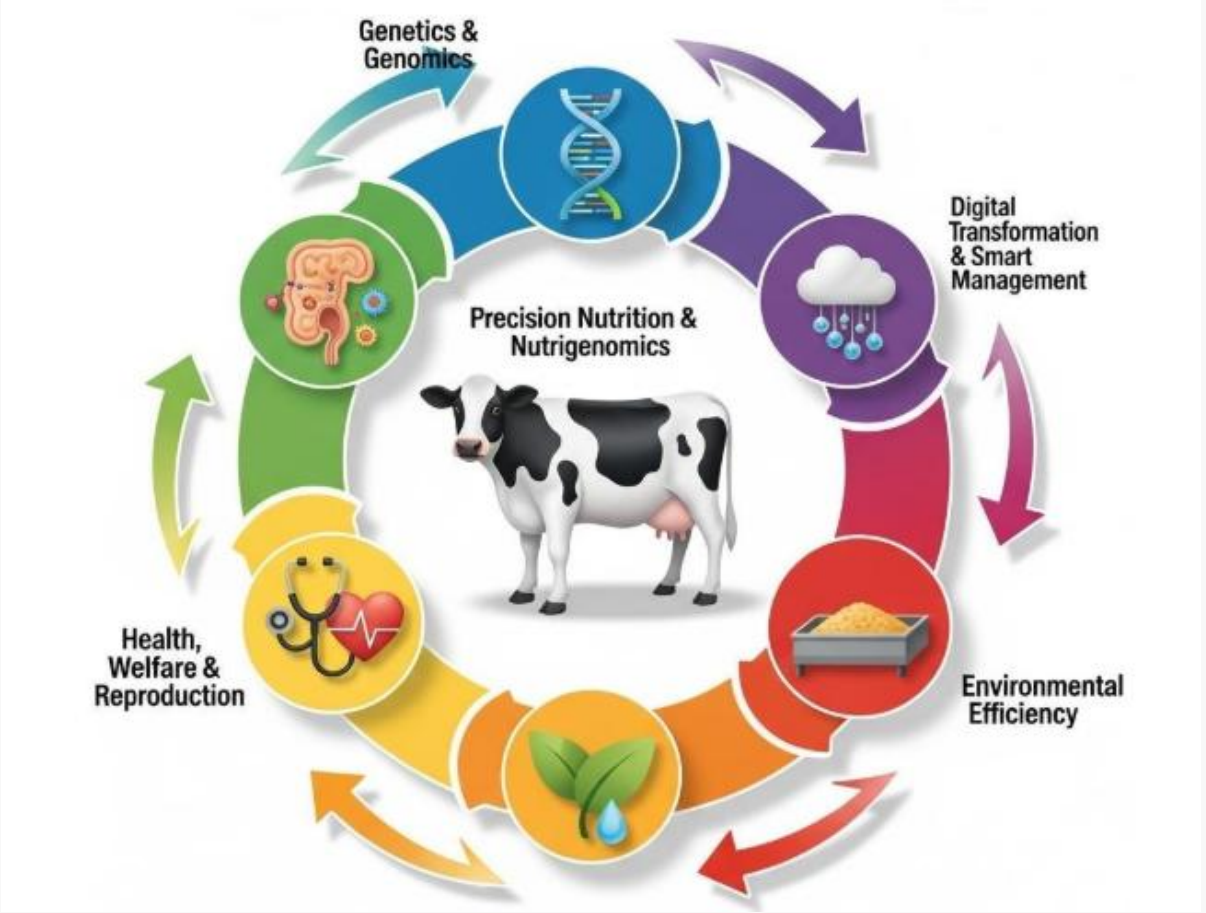
[OFFICIAL]

A resilience framework for tropical ruminant systems

Narrative

Murdoch University





[OFFICIAL]

Heat stress is the dominant entry point for resilience in the tropics

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Resilience is no longer defined only by the ability to cool cattle during acute events.

It increasingly means identifying animals and systems that maintain DMI, milk yield, fertility, and recovery under repeated thermal challenge.

Longitudinal phenotyping uses yield deviations, variability, and recovery as indicator traits.



[OFFICIAL]

Heat Stress & Adaptation

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At farm level, the field has converged on multi-layered mitigation:

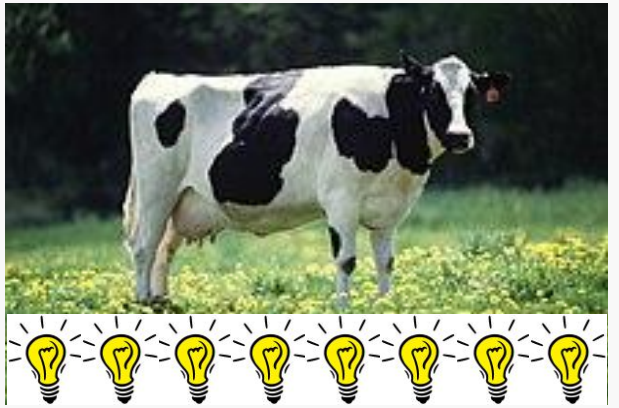
- shade, fans and sprinklers;
- ration adjustments that protect rumen function and water balance
- real-time monitoring of respiration, activity, temperature and milk responses.



[OFFICIAL]

Heat production





1 person (75 kg) generates about as much heat as a 100-watt light bulb

1 cow (650 kg) generates about as much heat as 8 100-watt light bulb

Lactating cow producing 40 L/day

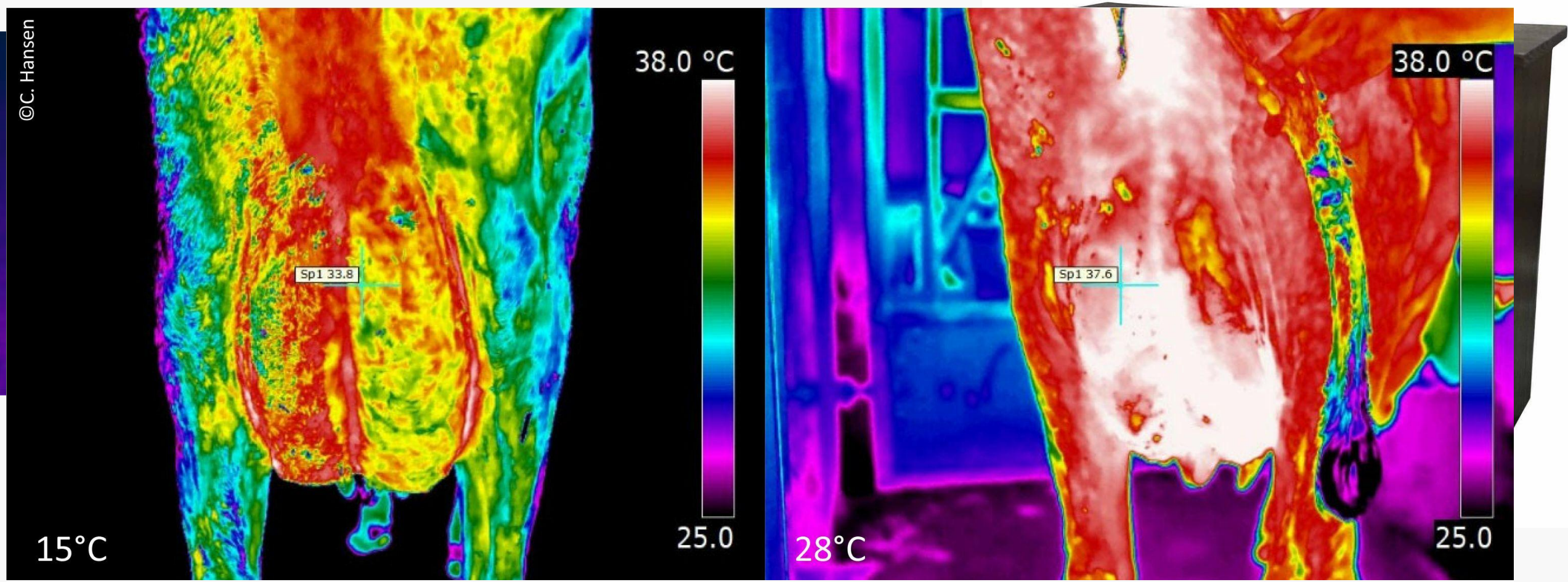
For milking cows, 5 liter of milk = to 1 light bulb





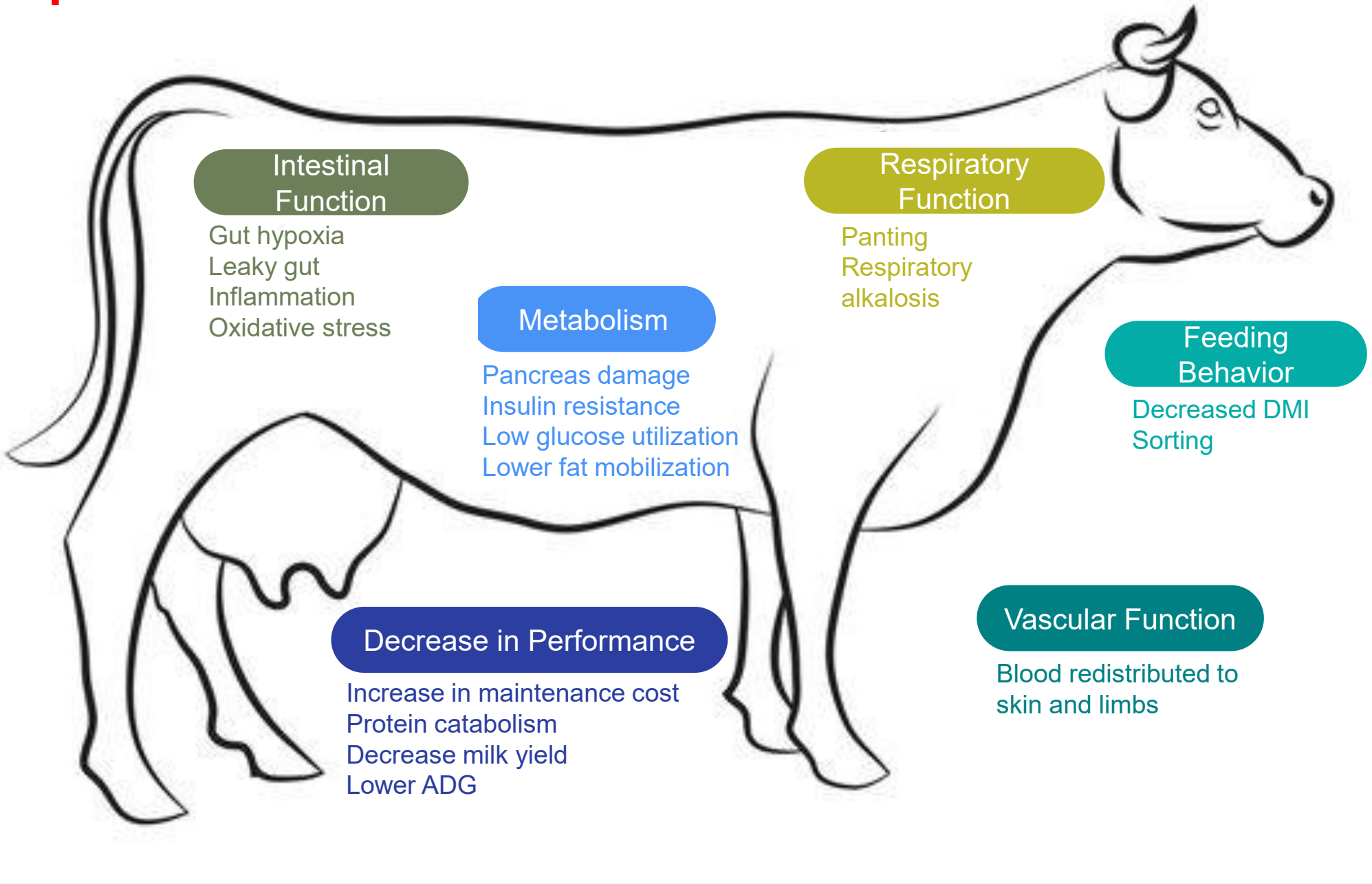
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Heat production



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Impact of heat stress on farm animals



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Heat Stress and Gut Health

Heat load disrupts gut integrity and amplifies systemic stress

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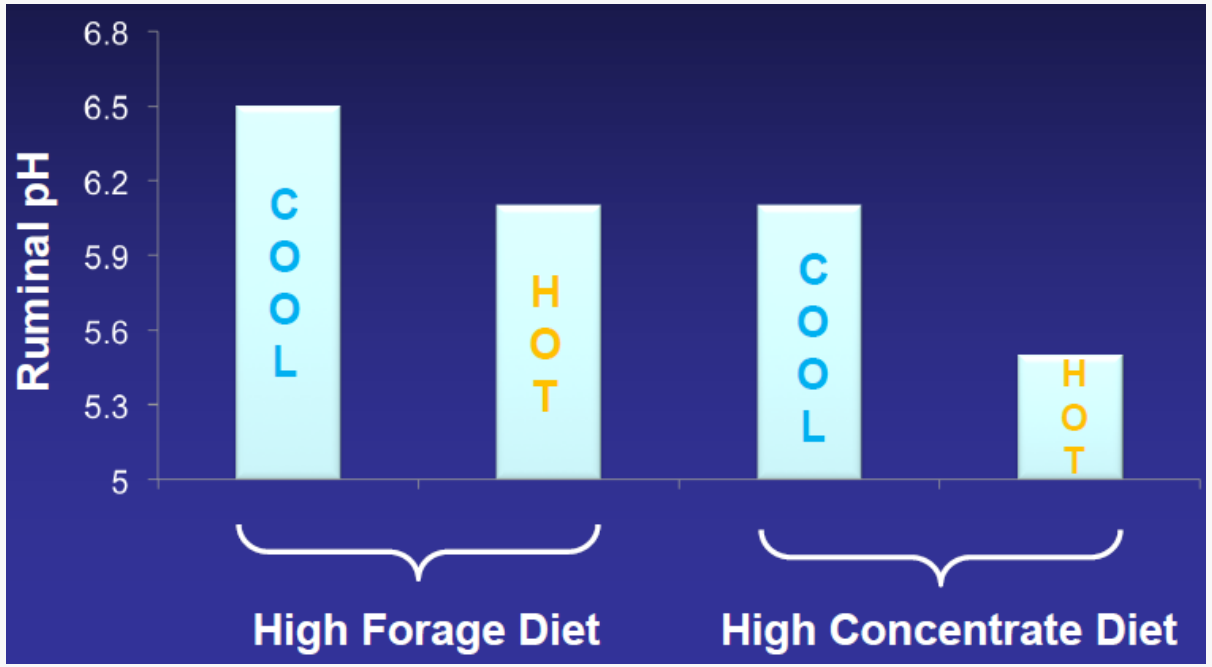
- Heat load
- Blood flow shift
- Gut hypoxia & leakiness
- Inflammation & endotoxin
- Lower intake and performance



[OFFICIAL]

Heat Stress Induced Rumen Acidosis

(Mishra et al., JAS 30:1023)



Changes in feeding patterns and diet composition

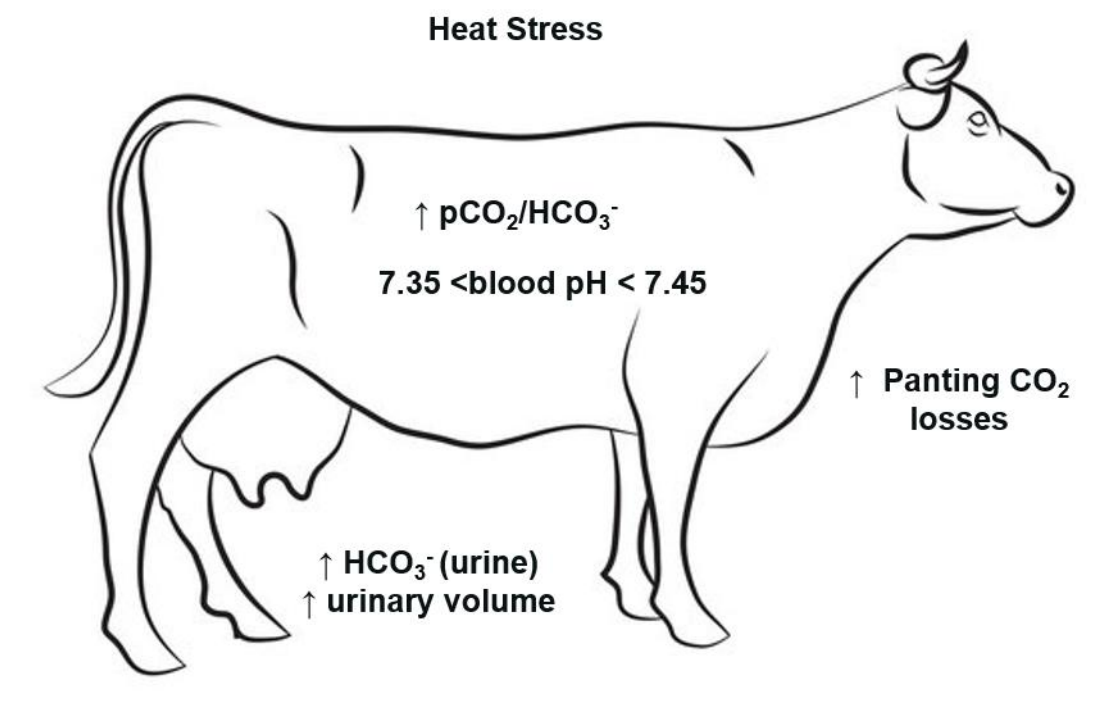
- Lower DMI.
- Higher proportion of concentrates in the diet.
- Feed sorting behaviour.
- Irregular or “slug” feeding episodes.
- Increased drooling.
- Reduced saliva production leading to further decrease in buffering capacity.

Baumgard and Rhoads, 2013

[OFFICIAL]

Heat Stress Induced Rumen Acidosis

(Mishra et al., JAS 30:1023)



Altered respiration patterns

- Leads to reduced systemic buffering capacity.

Hyperventilating causes blood alkalosis by decreasing pCO_2

To limit blood alkalosis, animal excretes HCO_3– in urine.

Less HCO_3– is available to maintain pH in the rumen.

[OFFICIAL]

Heat Stress and Gut Health

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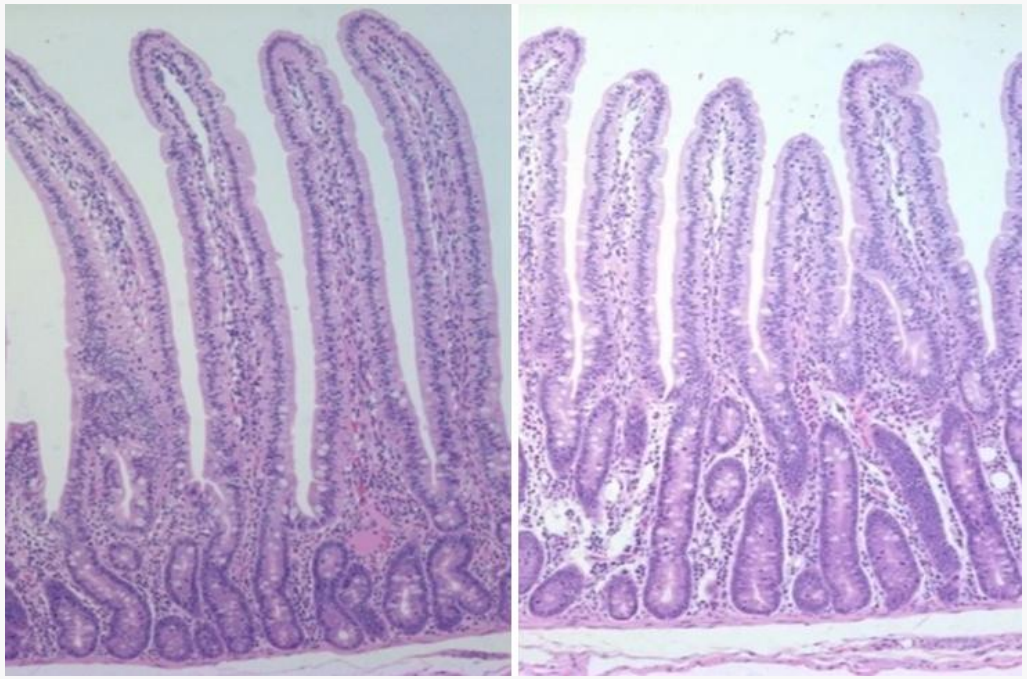
Diversion of blood flow to skin and extremities

Coordinated vasoconstriction in intestinal tissues

- Reduced nutrient and oxygen delivery to enterocytes
- Hypoxia increases reactive oxygen species (ROS)

Reduced nutrient uptake increases rumen and intestinal osmolarity in the intestinal lumen

- Multiple reasons for increased osmotic stress



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Heat Stress and Gut Integrity

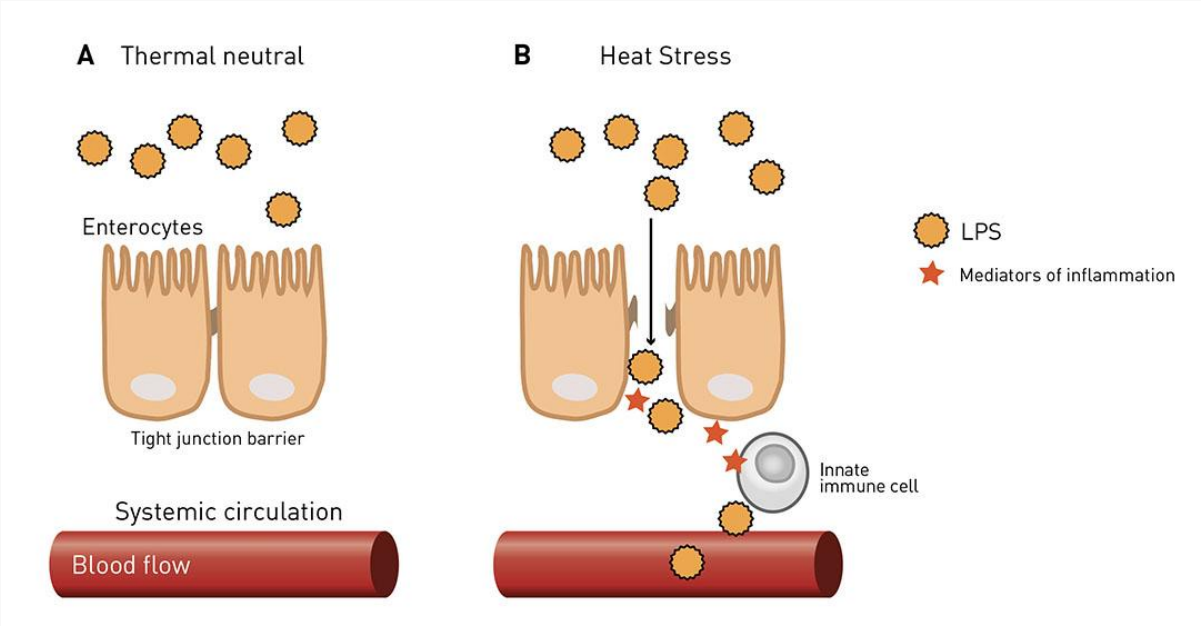
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Lipopolysaccharide (LPS) is a structural part of bacterial cell walls.

When bacteria die, LPS is released into the intestinal tract.

Under normal conditions, the gut's tight junctions prevent LPS from passing into the bloodstream.

During heat stress, gut barrier integrity is compromised, allowing LPS to leak into circulation.



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Heat Stress and Gut Integrity

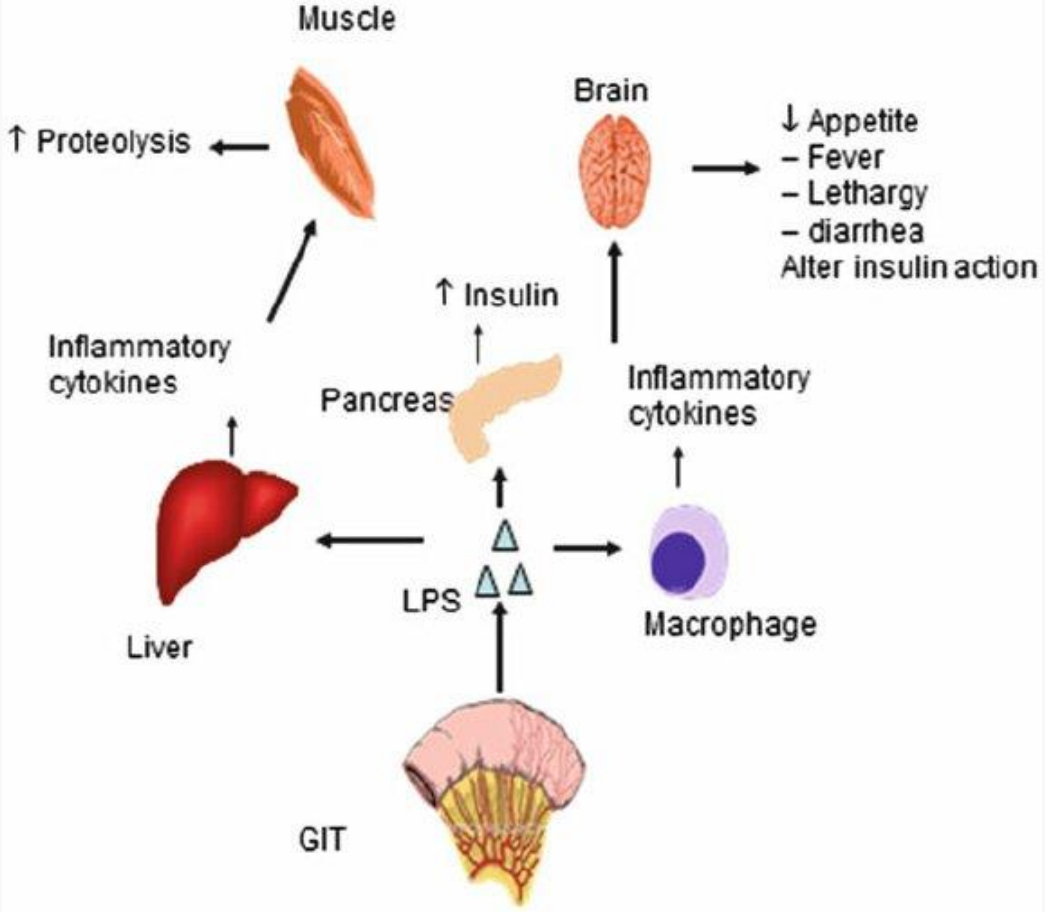
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LPS can damage the liver, leading to:

Reduced capacity for gluconeogenesis	Impaired export of VLDL, increasing risk of fatty liver	Lower secretion of anabolic hormones

Triggers inflammation through cytokines such as TNF-α and IL-1, creating a catabolic state:

Reduced feed intake	Elevated body temperature	Muscle breakdown	Reduced activity	Decline in productivity

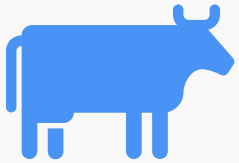


TNF-α: Tumor Necrosis Factor-alpha; IL-1: Interleukin-1; VLDL: very low-density lipoprotein.

[OFFICIAL]

Heat Stress and Glucose

A metabolic hallmark of heat stress is altered glucose allocation



Mechanistic summary

Heat-stressed cows fail to fully engage glucose-sparing responses.

Insulin action increases while non esterified fatty acids (NEFA) mobilisation declines, making glucose a more central fuel.

The result is inefficient energy partitioning and reduced support for lactation.



Liver and glucose

Adult ruminants depend heavily on hepatic gluconeogenesis for glucose supply, so disturbances in substrate flow and inflammatory burden can have system-wide consequences.



Implications for resilience

Limit heat load before severe metabolic compensation is required.

Use nutrition that supports stable DMI, rumen function and glucose sparing.

Interpret glucose metabolism as part of a broader resilience phenotype, not a stand-alone biochemical story.

[OFFICIAL]

Glucose metabolism under normal and heat-stressed conditions in cattle

Heat stress compromises both the availability and utilization of glucose

Heat-stressed cows fail to enlist glucose sparing mechanisms

Glucose appears to be the primary energy source possibly because of reduced lipolysis (less NEFA)

The result is

Inefficient energy metabolism

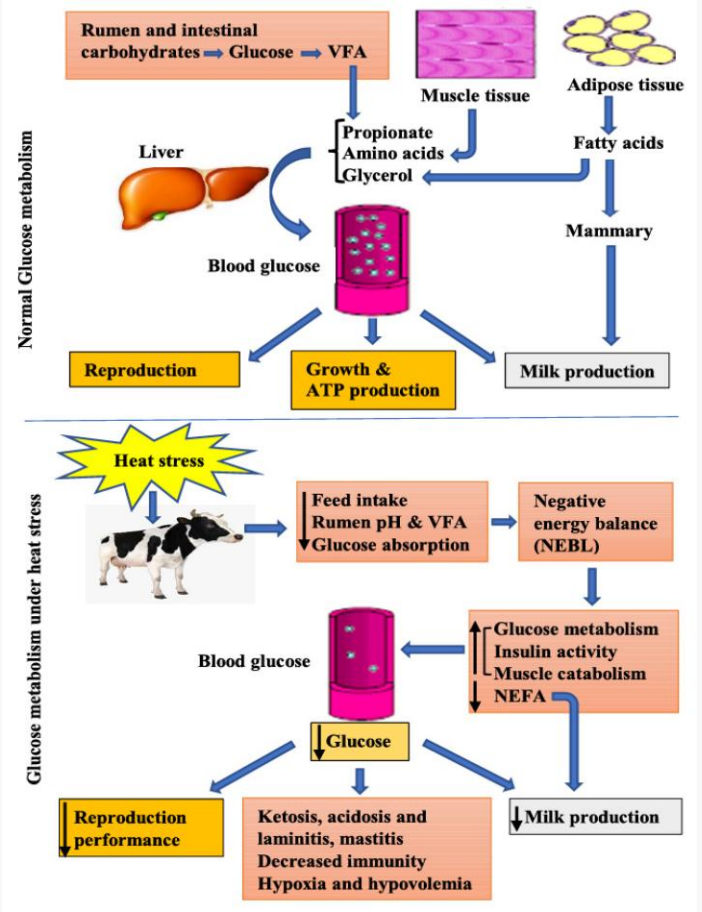
Reduced lactation performance.

This highlights the importance of nutritional and management interventions aimed at

Supporting glucose sparing,

Improving insulin sensitivity

Minimizing heat load



NEFA: non-esterified fatty acids

[OFFICIAL]

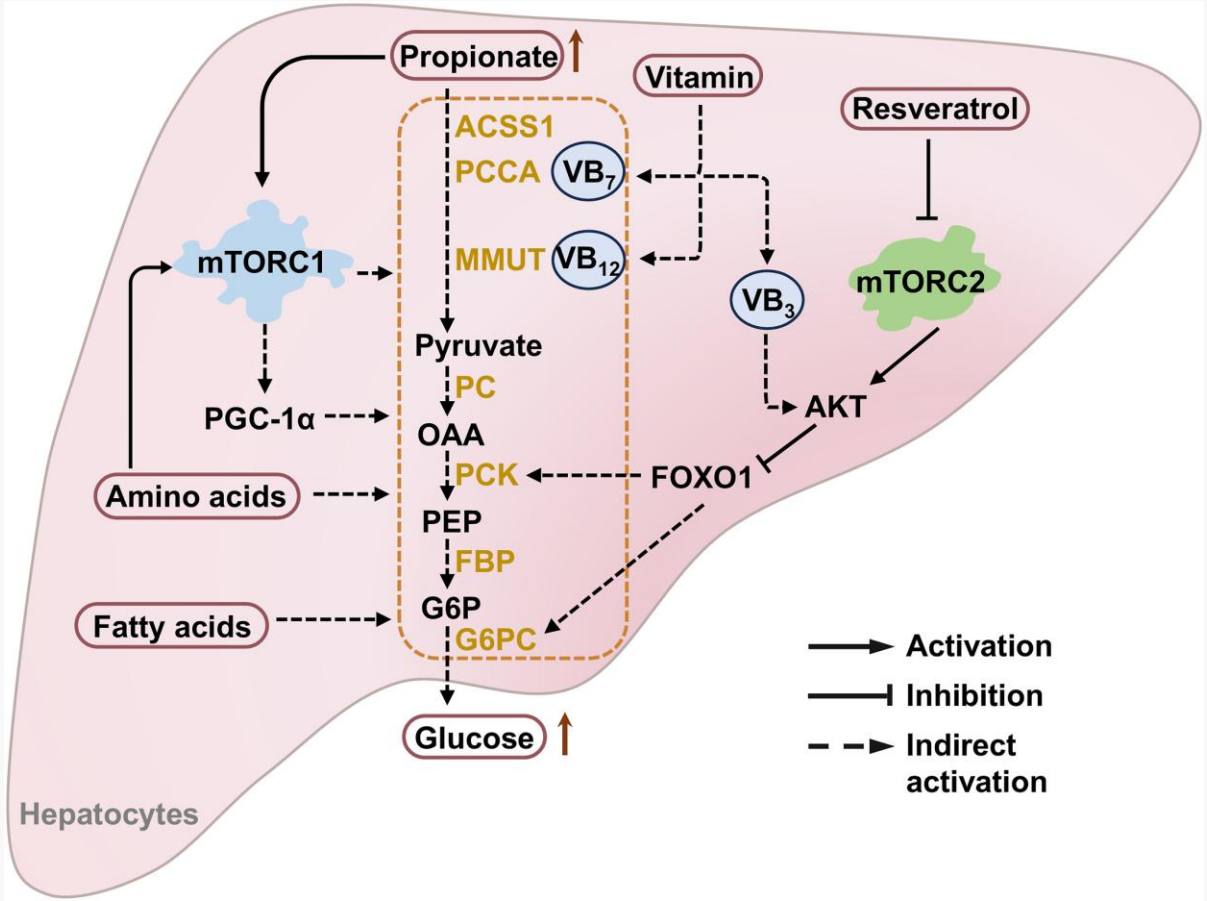
Glucose metabolism in dairy cows

Most of the glucose originates from the liver (glucose factory)

Gluconeogenesis in the liver is responsible for 90% of the total glucose needs in adult ruminants.

For example:

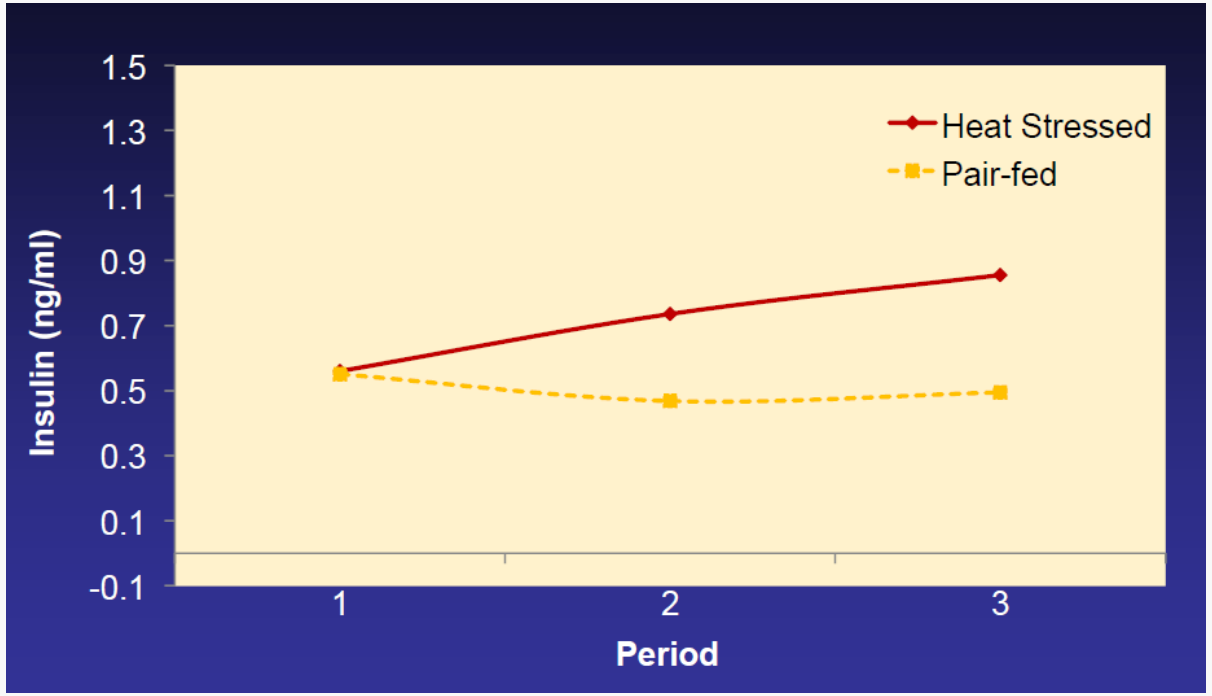
- a dairy cow producing 50 kg milk/day
- would require 3.6 kg of glucose
- 3.2 kg of which must come from gluconeogenesis



Wang et al 2024 https://doi.org/10.1002/aro2.80

[OFFICIAL]

Circulating Insulin in Cattle during heat stress



Heat-stressed cows have increased insulin action

Decreased NEFA

Increased glucose disposal

Heat-stressed cows require extra energy, especially glucose

Baumgard and Rhoads, 2013

[OFFICIAL]

Heat Stress mitigation pathway

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Heat load & oxidative stress and inflammatory strain

Phenotyping with milk yield, respiration rate, temperature, sensors

Mitigation via shade, cooling, ration and water management

Outcome: maintain DMI, fertility, welfare and milk yield



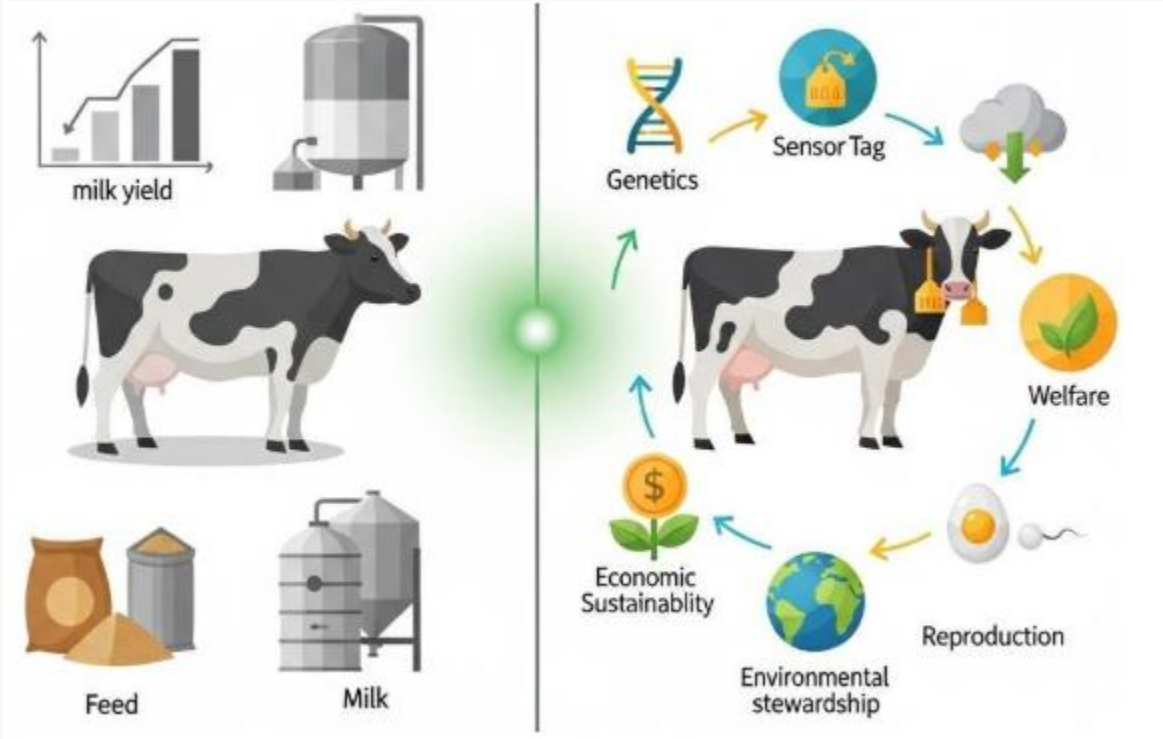
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Feed Efficiency

Feed Efficiency

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- Feed efficiency 1.1: less feed per litre of milk
- Feed efficiency 1.2: Nutrient-use efficiency
- Methane mitigation strategies integrated into feeding systems
- Metabolic stability
- Ability to cope with diet and climate variations
- Efficiency improvements linked to sustainability outcomes

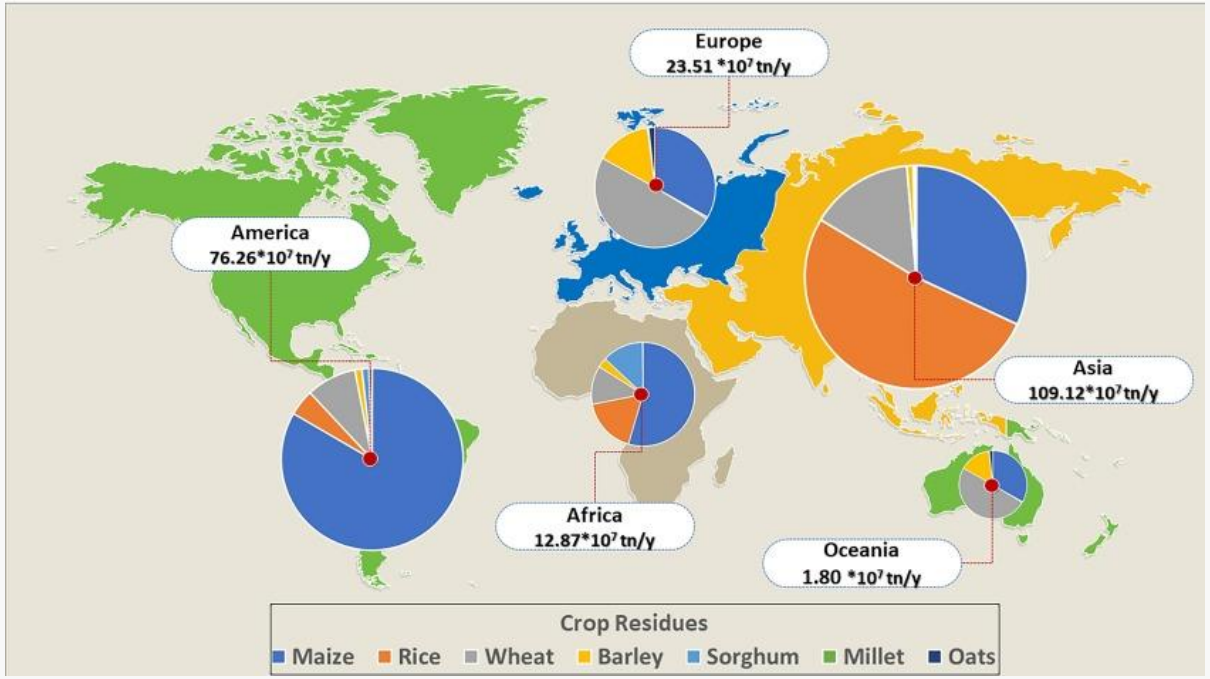


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Global importance of low-quality forages

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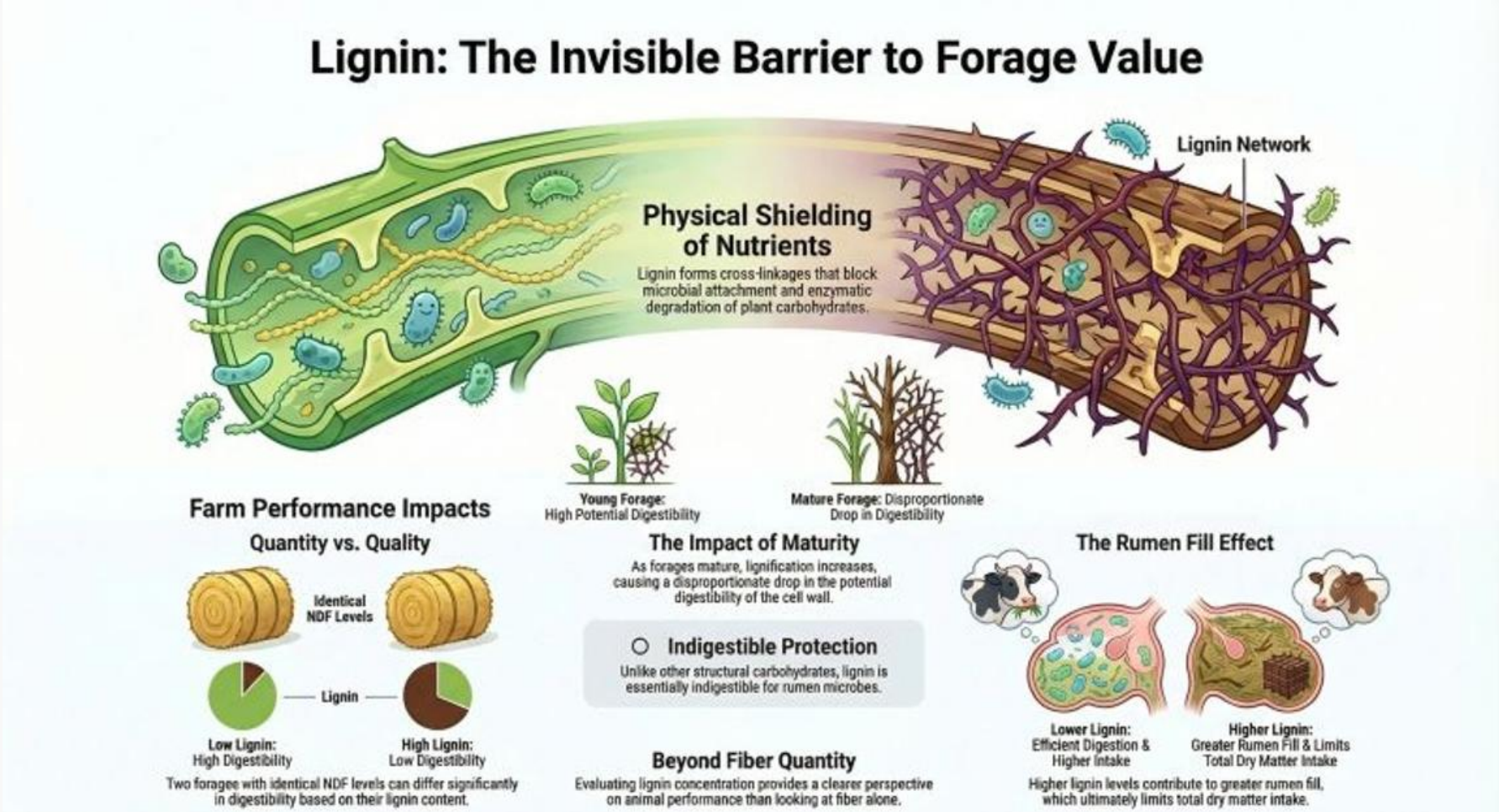
- Crop residues account for billions of tonnes annually
- Major feed source in tropical and subtropical systems
- Critical during dry seasons
- Digestibility limits animal productivity



[OFFICIAL]

What Defines a Low-Quality Forage?

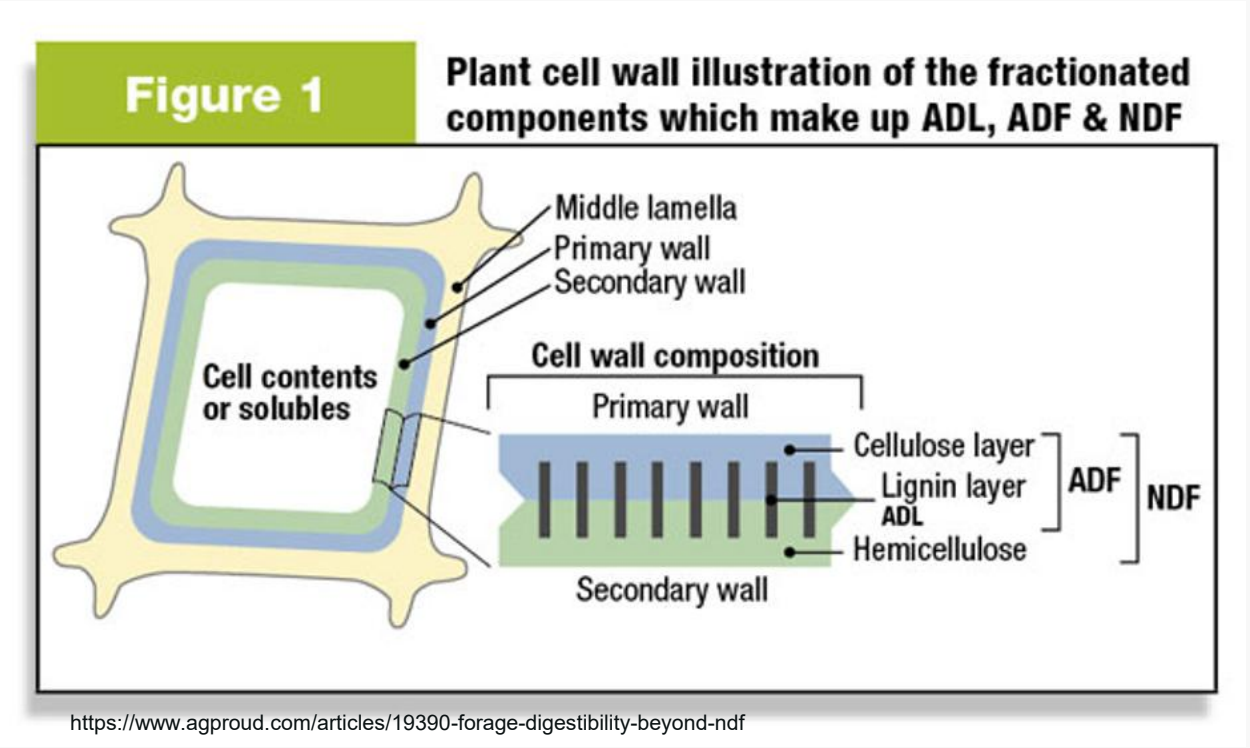
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increasing maturity → increasing lignin → decreasing digestibility.

[OFFICIAL]

Why does digestibility decline?



https://www.agproud.com/articles/19390-forage-digestibility-beyond-ndf

Increased lignification → Cellulose protected by lignin → Low ruminally available nitrogen → Reduced microbial colonisation

[OFFICIAL]

Low-quality forage: importance of energy

Energy is involved in many functions of dairy cows:

- Production, reproduction, health and immunity...

Energy is generally the first limiting factor for dairy cows, as it is much more difficult to increase the energy level of the diet than the protein level:

- Energy concentration of ingredients varies less than protein
- High levels of fat and starch lead to a global decrease of digestibility of the diet

Forage represents a major supply of energy of the diet





[OFFICIAL]

Low-quality forage: a major issue

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Inconsistent forage quality - Seasonal changes, plant maturity, and storage conditions lead to fluctuating nutrient profiles.

High fiber, low digestibility - Many forages contain high NDF and lignin, limiting microbial access.

Reduced rumen efficiency - Poor fiber degradation slows fermentation, lowers VFA production, and impacts animal performance.

Limited control over forage supply - Producers often rely on what's locally available, even if it's nutritionally suboptimal.



Forage quality effects digestibility and energy value

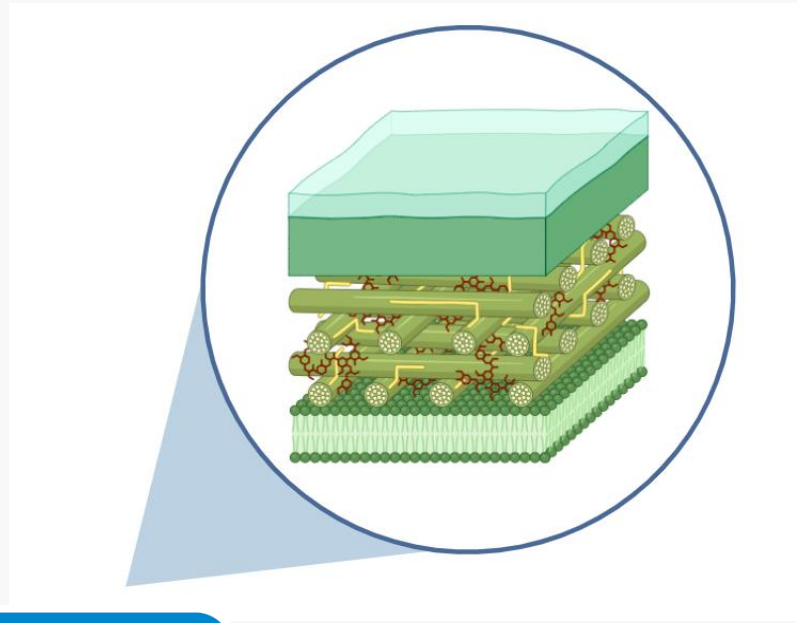
	NDF %	Lignin %	dNDF %[a]	Energy Value[b]
Forage A	44.9	3.34	60%	1.43
Forage B	43.5	2.43	68%	1.50

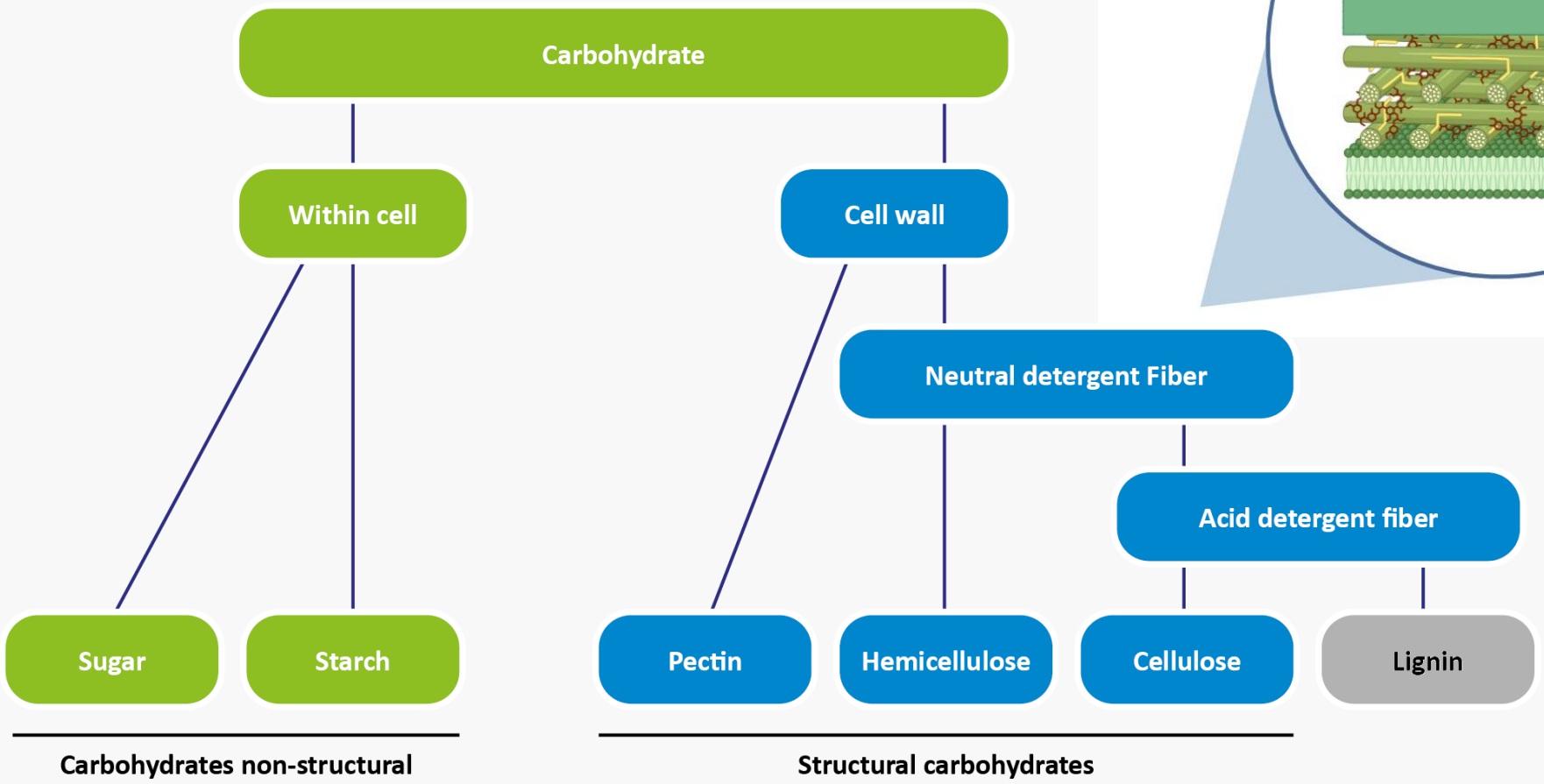
[a]Digestible Neutral Detergent Fiber

[b]NEL Mcal/kg

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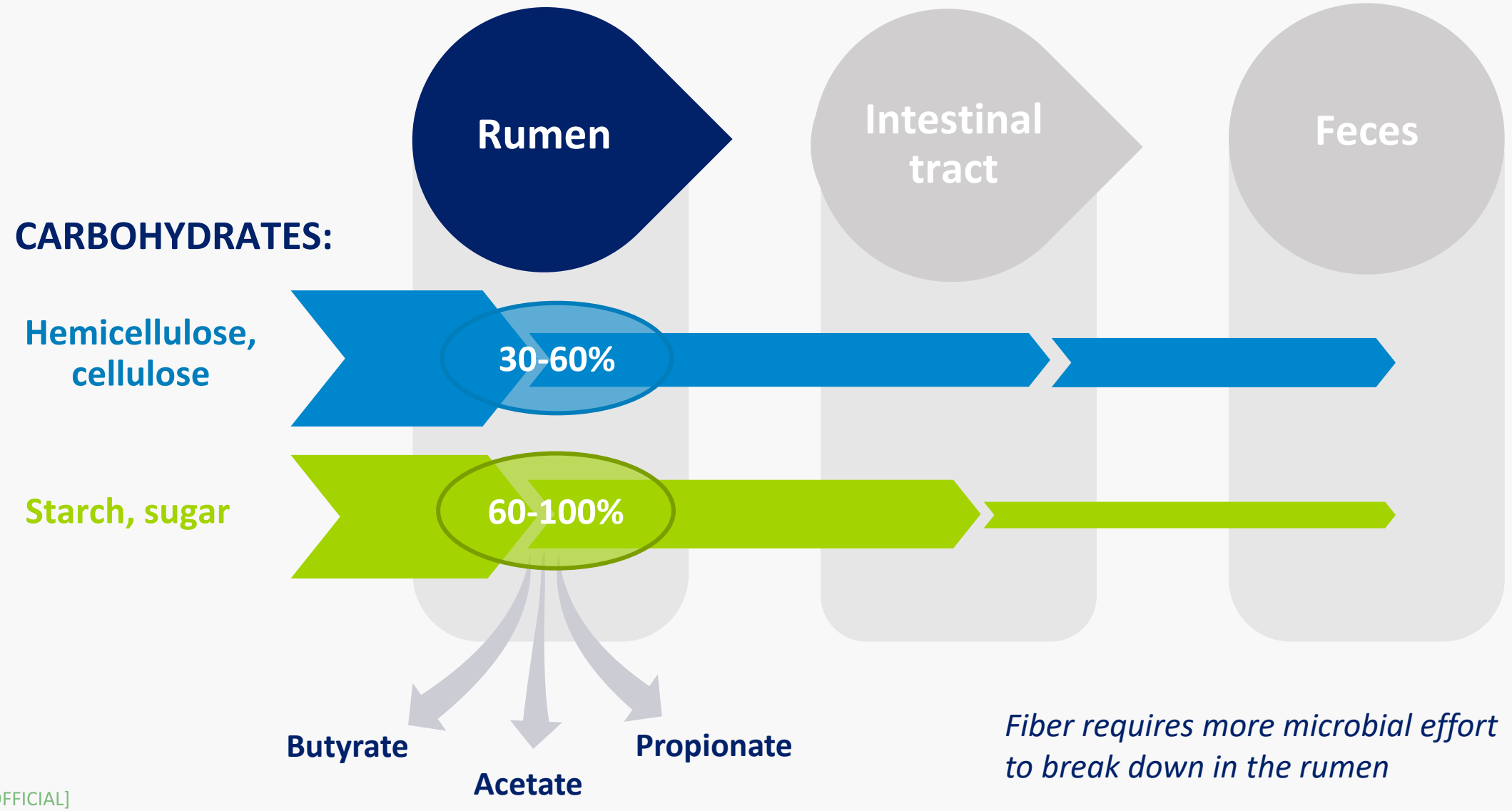
Energy efficiency – Carbohydrates definition





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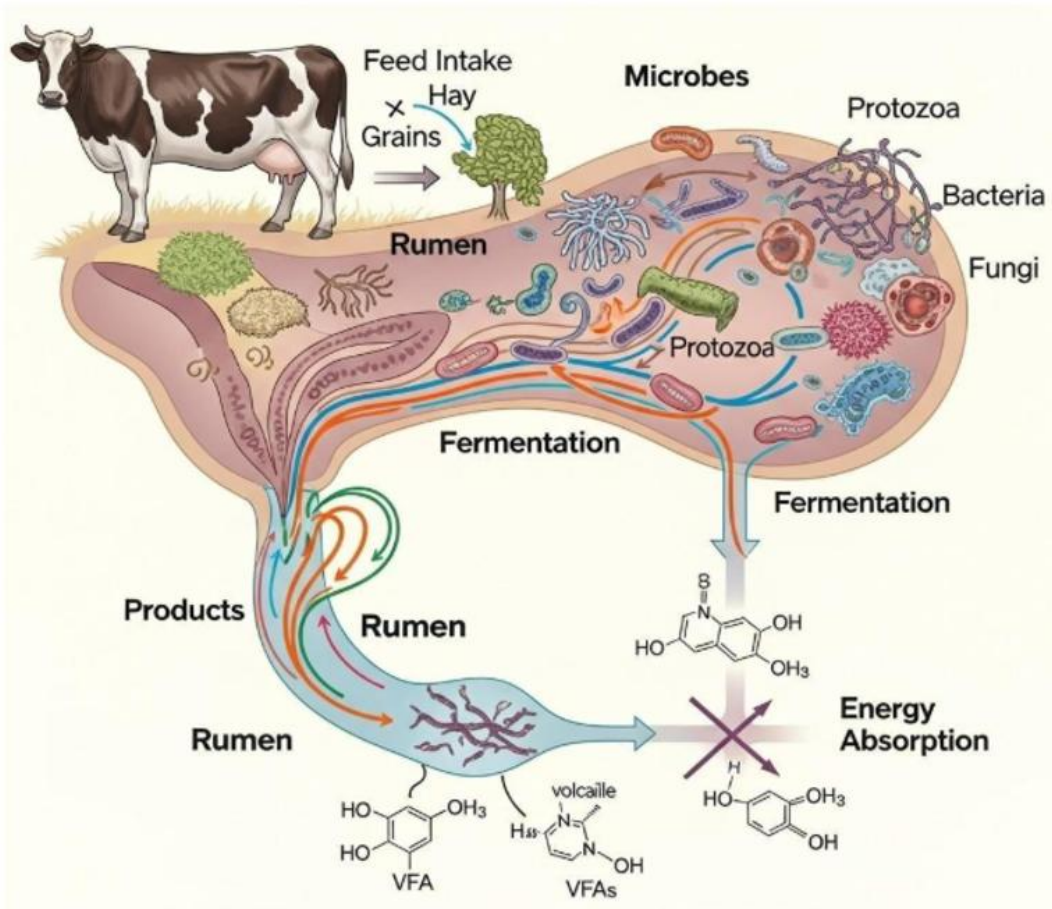
Carbohydrates digestibility



Fiber requires more microbial effort to break down in the rumen

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How to improve digestibility of carbohydrates?



Addition of exogenous microbes and enzymes

- live yeasts
- biotics
- amylase

Stimulation of existing microbes and enzymes

- minerals
- essential oils
- organic acids etc

Improved access to fiber for higher microbial attachment

- surfactants

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Conclusions

Practical considerations

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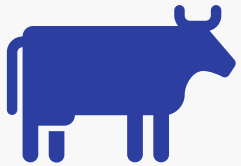


Production realities

Feed resource constraints and variable forage quality.

High heat and humidity with strong seasonal pressure.

Need for practical, field-applicable diagnostics and ration decisions.



Priority areas

Protect intake under heat stress.

Use locally available feeds more strategically.

Strengthen feed-risk and health monitoring.



Practical outcomes

Resilience concepts map directly onto themes: rumen function, metabolic efficiency, precision feeding, gut health, feed risk management, and field diagnostics.

This makes resilience a useful organising principle for both science and extension.

[OFFICIAL]

Key takeaways

Heat stress is the most visible disturbance, but its real importance lies in how it cascades through gut integrity, metabolism, intake, and reproduction.

Resilient nutrition combines precision feeding, rumen protection, and careful evaluation of methane-mitigation trade-offs.

For tropical systems, resilience should be treated as an integrated systems trait linking management, nutrition, health, and genetics.



[OFFICIAL]

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Thank you

Ngala kwop biddi.
Building a brighter
future, together.